Exponents & Exponential Functions Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Product rule

ORIGINAL PROBLEM

Simplify $x^3 \cdot x^4$

INCORRECT WORKED SOLUTION

$x^3 \cdot x^4$

$= x^{3 \cdot 4}$

$= x^{12}$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

2. Power of a product

ORIGINAL PROBLEM

Simplify $(2x^3)^2$

INCORRECT WORKED SOLUTION

$(2x^3)^2$

$= 2x^6$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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3. Quotient rule

ORIGINAL PROBLEM

Simplify $\frac{x^8}{x^2}$

INCORRECT WORKED SOLUTION

$\frac{x^8}{x^2}$

Divide the exponents: $\frac{8}{2} = 4$

$= x^4$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

4. Negative exponents

ORIGINAL PROBLEM

Evaluate 3^{-2}

INCORRECT WORKED SOLUTION

3^{-2}

$= -3^2$

$= -9$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

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5. Zero exponent

ORIGINAL PROBLEM

Simplify $4x^0$ $(x \neq 0)$

INCORRECT WORKED SOLUTION

$4x^0$

$= 4 \cdot 0$

$= 0$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

6. Negative base

ORIGINAL PROBLEM

Evaluate -3^2

INCORRECT WORKED SOLUTION

-3^2

$= (-3)(-3)$

$= 9$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

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7. Exponential growth

ORIGINAL PROBLEM

A culture of 200 bacteria doubles every hour. Write a model and find the amount after 3 hours.

INCORRECT WORKED SOLUTION

$y = 200 \cdot 2t$

After 3 hours: $y = 200 \cdot 2(3)$

$y = 200 \cdot 6 = 1200$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

8. Decay factor

ORIGINAL PROBLEM

A $20,000 car loses 15% of its value each year. Find its value after 2 years.

INCORRECT WORKED SOLUTION

$y = 20{,}000(0.15)^t$

After 2 years: $20{,}000(0.15)^2$

$= 20{,}000(0.0225) = \$450$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

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9. Multiplying terms

ORIGINAL PROBLEM

Simplify $(3x^2)(4x^3)$

INCORRECT WORKED SOLUTION

$(3x^2)(4x^3)$

$= 7x^5$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

10. Linear or exponential?

ORIGINAL PROBLEM

Table: x = 0, 1, 2, 3 and y = 3, 6, 12, 24. Write an equation.

INCORRECT WORKED SOLUTION

Differences: 3, 6, 12

The pattern grows, so $y = 3x + 3$

Check $x = 3$: $y = 3(3) + 3 = 12$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points